



NATIONAL INSTITUTE OF TECHNOLOGY, ROURKELA – 769008
END – SEMESTER EXAMINATION

SESSION: 2024 – 2025 (Spring)
B. Tech/Dual Degree/Int. MSc./M. Tech. / Ph.D

Subject code: CS 6301 **Sub:** Foundations of Computer Security
Full Marks: 50

Dept. Code: CS
Duration: 3 hrs

Answer all the questions. All parts of a question should be answered in one place.

Sl. No.	Questions	Marks	CO*	BL**
1.	a) What are the criteria of a cryptographic hash function? Explain each of them briefly.	[2]	CO1	BL2
	b) What is the value of Majority (p,q,r) where the values of p, q, and r are 0xB, 0x5, and 0xD respectively.	[2]	CO3	BL2
	c) What is the maximum message size, block size, message digest size, number of rounds and word size in SHA-1.	[2]	CO1	BL1
	d) Define Euler's Totient function.	[2]	CO2	BL1
	e) Write an algorithm in pseudocode to find QRs and QNRs for any Z_{p^*} .	[2]	CO2	BL1
2.	a) What is Elliptic Curve? Generate all points for the curve defined as $E_{11}(1,3)$. If (4,7) is a point on the curve then find $2 \cdot (4,7)$.	[5]	CO4	BL5
	b) Factor 97271 using the Pollard Rho method with polynomial $f(x) = x^2 + 1$ and initial guess $x_0 = 1$.	[5]	CO2	BL3
3.	a) Design a Polybius key matrix for play fair cipher using the keyword "STRONG" and encrypt the message " <i>the key is under the table</i> ".	[5]	CO3	BL3
	b) Explain the key generation, signing and verification phase of ElGamal Digital Signature Scheme with suitable diagram.	[5]	CO3	BL2
4.	a) Using Rabin cryptosystem with $p=11$ and $q=5$. I. Encrypt the plain text $P=13$ to find the cipher text. II. After receiving the cipher-text, use Chinese remainder theorem to find four possible plain texts.	[5]	CO3	BL3
	b) Explain Miyaguchi-Preneel Scheme with block diagram and discuss one cryptographic hash function based on this scheme.	[5]	CO2	BL2

P.T.O

Sl. No.	Questions	Marks	CO*	BL**
5.	a) Explain the working principle of Knapsack crypto-system. Consider, a super-increasing sequence 'w' as {3, 8, 12, 24, 48, 99, 198, 394}. Choose the modulus n as 881, r as 37, and [6 2 3 1 4 7 5] as permutation table. Suppose sender wants to send a message 'd' (Hint: use 7 bit ASCII value of a). Find the cipher-text and also write steps to verify the message thus received. b) Briefly explain the idea behind the RSA cryptosystem. I. What is the one-way function in this system? II. What is the trapdoor in this system? III. Define the public and private keys in this system. IV. Describe the security of this system	[5] [5]	CO4 CO3	BL4 BL1

***CO: Course Outcome; **BL: Bloom's Taxonomy Level**